Digital Transformation — Module 01: Systems — Discussion Questions

20 questions on sociotechnical systems, digital change levels, and organizational technology architecture.

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1. Why is digital transformation fundamentally a system design problem rather than a technology adoption problem?
2. How does Amazon's algorithmic organization illustrate the concept of technology as Markov blanket?
3. What distinguishes digitization, digitalization, and digital transformation? Give examples of each.
4. How does technical debt function as "accumulated model error" in organizational infrastructure?
5. When does technology constitute the organizational boundary versus merely supporting it?
6. How should the design of human-technology interfaces change based on the complexity of the task?
7. What determines whether a legacy system is an asset (institutional memory) or a liability (technical debt)?
8. How do microservices architectures change the system dynamics of digital organizations?
9. Compare the sociotechnical design challenges of a "born digital" company versus a traditional company undergoing digital transformation.
10. How should organizations balance investing in new technology versus maintaining existing systems?
11. Map the sociotechnical system of your organization. Where are the human-technology integration points? Where are the gaps?
12. Identify three areas where technology serves as your organizational boundary. How effective is it?

13. Assess your organization's technical debt. Where is the accumulated model error highest?
14. Design a sociotechnical system for a specific business process, specifying human and technology roles.
15. When should an organization choose a "big bang" technology replacement versus incremental migration?
16. How do cloud computing and SaaS change the system architecture of organizations?
17. What happens when the technology boundary fails (system outage)? How does this compare to human boundary failure?
18. How does the pace of technology change affect organizational system stability?
19. Compare how a hospital, a bank, and a tech startup should approach sociotechnical system design differently.
20. Design a "system health dashboard" that monitors both the human and technical subsystems of your organization.